Fixed prefixes and support deficits at a unique orbit crossing

Human Compression project: L044 working continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be a faithful transitive string action by involutions. Suppose colour f has a unique edge $\{a, b\}$ whose endpoints are in different orbits of $G_f = \langle \rho_j : j \neq f \rangle$. The single-crossing lemma in *Paired orbit cuts in string actions* gives exactly two G_f-orbits A, B, containing a, b, and says that every ρ_j with $|j - f| > 1$ fixes both a, b. Write $G_{>l} = \langle \rho_j : j > l \rangle$ and define $G_{<l}$ dually.

Lemma 1 (A fixed prefix to an active colour)**.** *Suppose $l > f$ and ρ_l acts nontrivially on A. There is a path in the full permutation graph from a to a point moved by ρ_l, using only colours less than l and different from f. Every vertex of this path is fixed by $G_{>l}$. The dual assertion for $l < f$ uses only colours greater than l and different from f, and its vertices are fixed by $G_{<l}$. The same assertions hold with b, B in place of a, A.*

Proof. Put $H = \langle \rho_j : j < l,\ j \neq f \rangle$ and $R = G_{>l}$. Every generator of R fixes a, because $j > l > f$ implies $j - f \geq 2$. Every generator of R also commutes with every generator of H. Consequently R fixes the whole orbit a^H pointwise. If ρ_l fixed that orbit pointwise as well, then the orbit would be invariant under all the generators of G_f. It would therefore equal $A = a^{G_f}$, contradicting the assumed nontrivial action of ρ_l on A. Some H-word thus takes a to a point moved by ρ_l. Its successive vertices lie in a^H and are fixed by R. Erasing loops gives the required simple path. Reversing the index order proves the dual claim; the argument at b is identical. □

Lemma 2 (Consecutive-colour lower bound)**.** *Under the hypotheses of Lemma 1 with $l > f$, every path from a to a point moved by ρ_l that avoids colour f contains each of $f + 1, \dots, l - 1$. In particular appending the final l-edge to a shortest prefix in Lemma 1 gives a simple path with at least $l - f$ edges. Its prefix, excluding that final edge, is pointwise fixed by $G_{>l}$.*

Proof. If a colour k with $f < k < l$ were missing, factor the path word into a word in generators of index less than k and a word in generators of index greater than k. The two sets commute. Every generator of index greater than k fixes a, because its index is at least $f + 2$. Thus the endpoint is reached from a by the lower-index word alone. The generator ρ_l commutes with that word and fixes a, so fixes the endpoint, a contradiction. This proves the colour and length assertions. Choose a shortest path, among the permitted prefixes of Lemma 1, to any point moved by ρ_l. Every earlier vertex is then fixed by ρ_l. The other endpoint of the final l-edge is also moved by ρ_l, so cannot be an earlier prefix vertex. The extended path is simple. □

Lemma 3 (The endpoint case of source Proposition 5.19)**.** *Suppose $f = 0$, every generator is even, $r \geq 4$, and ρ_{r-1} acts nontrivially on both G_0-orbits. Then $n \geq 2r + 1$, equivalently $r \leq \lfloor (n - 1)/2 \rfloor$.*

Proof. Apply Lemma 2 in each orbit with $l = r - 1$. The two disjoint simple paths each have at least $r - 1$ edges, so together they have at least $2r$ vertices. It remains to exclude $n = 2r$.

At equality the paths exhaust their respective orbits. Each has exactly one edge of each colour $1, \dots, r - 1$, in that order: the no-gap assertion applied to each initial segment forces the first occurrence of every colour to follow all smaller positive colours. Write the paths

$$a_0 = a -_1 a_1 -_2 \cdots -_{r-1} a_{r-1}, \qquad b_0 = b -_1 b_1 -_2 \cdots -_{r-1} b_{r-1}.$$

Because a_0b_0 is the unique cross-orbit 0-edge, ρ_0 preserves each tail $A' = \{a_1, \ldots, a_{r-1}\}$ and $B' = \{b_1, \ldots, b_{r-1}\}$. The group $H = \langle \rho_2, \ldots, \rho_{r-1} \rangle$ fixes a_0, b_0 and preserves both tails; its action on each tail is transitive, since the displayed edges form a path through that tail.

The subgroup $K = \langle \rho_3, \ldots, \rho_{r-1} \rangle$ fixes a_1: each of its generators fixes a_0 and commutes with ρ_1. Every other vertex of A' is moved by one of the displayed edges of colour at least three. This uses $r \geq 4$, and shows that a_1 is the unique K-fixed point in A'. The restriction of ρ_0 centralises H, hence K, so must fix a_1. Centralising the transitive H-action then forces it to fix every point of A'. The identical argument fixes all of B'. Therefore $\rho_0 = (a_0\, b_0)$ is a single transposition, contrary to evenness. This excludes equality. □

For the main L044 range $n \geq 12$, ranks $r \leq 3$ already satisfy the required bound, so Lemma 3 covers the whole endpoint/top-colour case. The argument does not assume full IP, does not use the unresolved square-rigidity assertion, and does not assert that all other source branches are complete.

Lemma 4 (The side of a shared-colour tail)**.** *Suppose $f = 0$, the noncommutation diagram of the given marking is connected, and $h < r - 1$ is the largest index whose generator acts nontrivially on both A and B. All generators of index greater than h act nontrivially on just one and the same orbit, say B. In particular $G_{>h}$ fixes A pointwise.*

Proof. Each generator of index greater than h is nonidentity and acts on at most one of the two G_0-orbits. It therefore acts on exactly one. If consecutive generators in this tail acted on different orbits, their supports would be disjoint, so they would commute. That would disconnect the noncommutation diagram: all nonconsecutive generators already commute by the string condition. Thus the entire tail acts on the same orbit. □

Lemma 5 (A support deficit without the source orbit reversal)**.** *Under the hypotheses of Lemma 4, suppose $h \geq 1$. Put $X = \mathrm{supp}(G_{>h})$. Then*

$$X \subseteq B, \qquad |\operatorname{Fix}(G_{>h})| \geq 2h + 1, \qquad h \leq \frac{n - |X| - 1}{2}.$$

More precisely, the fixed set contains the disjoint union of an A-path from a including its final h-edge, with at least $h+1$ vertices, and the prefix of a B-path from b to an h-edge, with at least h vertices. If equality holds, these paths have exactly the indicated sizes and exhaust the fixed set; A is the first path's vertex set. The paths are ordered by the colours $1, \ldots, h$. This last assertion is a description of the specified paths and fixed set, not a classification of all full-graph edges.

Proof. The A-path supplied by Lemma 2, including its last h-edge, is fixed by the whole tail because that tail fixes all of A. It has at least $h + 1$ vertices. The B-prefix is fixed by Lemma 1, and has at least h vertices. These sets are disjoint, proving the bound. At equality the two lower bounds are exact and account for every fixed vertex. Since every point of A is fixed, no other point of A can remain. The first occurrence/no-gap argument used in Lemma 3 orders the path colours. There is no claim that the final B-endpoint is fixed: its omission is precisely why the count is $2h + 1$. □

For the L044 application, a nontrivial faithful involutory marking of the simple group A_n ($n \geq 5$) has connected noncommutation diagram. Otherwise two nonempty sets of nonidentity generators would generate commuting normal factors of A_n; simplicity and its trivial centre exclude that. Thus Lemma 4's explicit hypothesis is available. No C-group intersection argument is needed for these support bounds.

The main inequality and support placement in Lemma 5 give the useful part of source Proposition 5.20. Its literal statement places X in the orbit *fixed* by ρ_{r-1}, while its proof defines X as the support of the tail containing ρ_{r-1}. The actual side is B, on which that generator acts. The source's displayed whole permutation-graph assertion has not been replaced by an

unproved claim: only the exact path/fixed-set consequence above is established here. Its later use must preserve that boundary.

Lemma 6 (Exact exceptional end under an alternating tail)**.** *In Lemma 5, suppose the marking satisfies IP, all its generators are even, $G_{>h}$ is the full natural alternating group on X, and $|\operatorname{Fix}(G_{>h})| = 2h+1$. Then $h = 2$, and $G_{<3}$ is exactly the six-point marking in source Table 2(2), fixing every other point.*

Proof. The nontrivial alternating group generated by the involutions of the tail acts on at least five points: A_3 is odd-order and the involutions of A_4 generate only its Klein four subgroup. The centraliser of the natural A_X in $\mathrm{Sym}(X)$ is trivial. For example its point stabiliser fixes only that point, so an element of the centraliser fixes one point and then all points by transitivity. Since ρ_0 commutes with the tail, it fixes X pointwise.

Write the equality paths as in Lemma 3, stopping at index h. The endpoint b_h lies in X: the rest of the two paths already exhaust the fixed set. Thus ρ_0 fixes b_h. Commutation with $\rho_h, \rho_{h-1}, \ldots, \rho_2$ propagates this fixation to $b_{h-1}, \ldots, b_1$. Hence its only possible remaining action, besides $(a_0\ b_0)$, is on $A' = \{a_1, \ldots, a_h\}$.

If $h = 1$, that set is a singleton, contradicting evenness of ρ_0. If $h \geq 3$, the group $\langle \rho_2, \ldots, \rho_h \rangle$ is transitive on A', and its subgroup $\langle \rho_3, \ldots, \rho_h \rangle$ has exactly one fixed point there, namely a_1. The proof is the same displayed-path argument as in Lemma 3. Centralising both groups forces ρ_0 to fix A' pointwise, again a contradiction. Therefore $h = 2$ and evenness gives

$$\rho_0 = (a_0\ b_0)(a_1\ a_2).$$

Now ρ_1 centralises the whole tail, so fixes X pointwise. On the three-point orbit A and the two fixed points of B it is therefore exactly

$$\rho_1 = (a_0\ a_1)(b_0\ b_1).$$

The generator ρ_2 fixes a_0, b_0 and has the form

$$\rho_2 = (a_1\ a_2)(b_1\ x)\delta, \qquad x = b_2 \in X,$$

where δ is an even involution supported on $X \setminus \{x\}$. On the six displayed points the three core permutations generate the natural degree-six action of A_5; the complete small certificate `small_tail_model.py` identifies them, generator by generator, with three permutations in $A_5 \leq S_5$. This is the source's Table 2(2), with its vertices read left to right as $a_2, a_1, a_0, b_0, b_1, x$.

If $\delta \neq 1$, the prefix $G_{<3}$ is a subgroup of the direct product of this core A_5 and $\langle \delta \rangle$, projecting onto both factors. If it did not contain δ, it would be the graph of a homomorphism $A_5 \to C_2$ taking the third core involution to the nonidentity element. Perfection of A_5 excludes such a homomorphism. Thus $\delta \in G_{<3}$; but $\delta \in A_X = G_{>2}$ as well, contrary to IP for the disjoint index sets $\{0, 1, 2\}$ and $\{3, \ldots, r-1\}$. Consequently $\delta = 1$, proving the exact marked conclusion. □

This lemma supplies the endpoint alternating-tail equality obstruction used in the source's Proposition 5.26. It does not replace the other prefix cases, the full Table 2 census, or the remaining single-crossing induction.

Lemma 7 (A transitive tail with an earlier commuting involution)**.** *Assume every maximal omission subgroup G_k is intransitive. Let $H = G_{>j}$ be transitive on its support X, and suppose ρ_l, for some $l < j$, acts nontrivially on X. Then*

$$r - 1 - j \leq |X|/2 - 1.$$

No IP is required for this inequality. It uses the retained Whiston bound for an independent family in a symmetric group.

Proof. The restriction $z = \rho_l|_X$ commutes with H. An element centralising a transitive group and fixing one point fixes all points. Hence the nontrivial involution z has no fixed point, and its pairs give an H-invariant block system with $m = |X|/2$ blocks.

Choose a minimal subset L of the displayed H generators whose images generate the whole block action. The group $\langle L\rangle$ is transitive on these blocks and has either one or two orbits on X. If there are two, z interchanges them (each meets every pair in one point). Since H itself is transitive, there is a generator ρ_k of H outside L. Then G_k contains both $\langle L\rangle$ and ρ_l, and is transitive on X. If instead $\langle L\rangle$ is already transitive, G_k is transitive on X for any H generator ρ_k outside L.

In either case an omitted ρ_k is supported entirely in the single G_k-orbit containing X. Adjoining it cannot join any G_k-orbits. Since $\langle G_k, \rho_k\rangle = G$ is transitive, G_k would already be transitive, a contradiction. Thus every H generator belongs to L. Their block images form an independent family in S_m, so their number is at most $m - 1$ by Whiston. □

Lemma 7 supplies the rank inequality of source Proposition 5.21. Its displayed equality graph classification is a separate obligation; it is not inferred from Whiston's cardinality bound alone.

Exact relation to the source. The no-gap idea comes from Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. The relevant source assertion is Proposition 5.18, on physical pages 29–30. The statement above specifies the initial orbit, the active target colour, the side of the string which fixes the path, and the exclusion of its final edge. The source's final sentence instead prints $G_{>l}$ when $l < f$ and $G_{<l}$ when $l > f$ and does not make the prefix boundary explicit. We do not identify that literal sentence with Lemma 1. No bound for the whole original, new conjecture solution or formal verification is asserted here.